

Paarth Patodia

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EDUCATION

University of Massachusetts Amherst

B.S. Electrical Engineering

Expected May 2028

GPA: 3.589/4.0

- **Honors:** Chancellor's Award, Dean's List
- **Relevant Coursework:** Continuous-Time Signals & Systems, Circuits and Electronics I, Intro to Embedded Systems, Modern Physics & Materials for EE, Probability and Statistics, Analytical Tools for ECE, Computational Tools for ECE, Multivariate Calculus

SKILLS

Languages & Tools: Python (NumPy, SciPy, pandas, Matplotlib, Plotly, Streamlit, pytest, openpyxl), Java, MATLAB (coursework), Git, JSON, Excel

Battery & Power Systems: DCIR, HPPC and capacity test analysis; coulomb-counted SOC; OCV and voltage hysteresis extraction; 1RC–5RC Thevenin equivalent-circuit modeling; nonlinear least-squares parameter identification; model validation for BMS cell modeling

Embedded & Hardware: Arduino, PWM control firmware, MOSFET driver and switching circuits, ESP32, OLED interfacing, breadboard prototyping, soldering, FLIR thermal imaging validation

Test & Automation: Selenium, TestNG, Maven, Page Object Model (read, debug and modify); n8n, MCP, REST APIs, webhooks, LLM agent orchestration

Certification: Stanford Algorithms Specialization (Coursera)

EXPERIENCE

Maruti Suzuki

Electrical Engineering Intern — Battery Research / Battery Technology

Summer 2026

India

- Developed a Python-based battery characterization analytics framework automating DCIR, HPPC and capacity test analysis across 2 suppliers (3 cells each), processing 100,000+ pulse events and generating 30+ engineering comparison plots, reducing manual analysis from hours to minutes.
- Performed comparative analysis of lithium-ion (LFP) characterization data across SOC, temperature and C-rate to evaluate DCIR, power capability and cell consistency, supporting engineering decisions for EV battery cell selection.
- Built an automated parameter estimation framework for 1RC and 2RC equivalent circuit models, using bounded nonlinear least squares (scipy.least_squares, trust-region reflective) to identify model parameters from experimental data.
- Applied DCIR and HPPC data to ECM parameter identification, model validation and simulation correlation, improving agreement between simulated and experimental voltage responses for Battery Management System cell modeling.

Leapedge Aerospace and Defense Technologies Pvt Ltd

Embedded Systems Engineering Intern

Jan 2026

Sohna, India

- Built a prototype infrared target simulator using a resistive heater array to replicate thermal signatures for FLIR camera testing on a defense imaging project.
- Implemented PWM-based heater control firmware on Arduino to drive and switch heating patterns across the array.
- Assembled and tested MOSFET-driven heater channels under senior-engineer guidance, and validated programmed heating patterns against captured thermal camera output.

MCode

AI Developer Intern

Feb 2026 – May 2026

India

- Developed and tested modules within AI agent and automation workflows across four client engagements, connecting LLMs to enterprise data, APIs and operational systems using Python, Claude, n8n and MCP.
- Built the n8n orchestration and data-processing layer for a natural-language manufacturing intelligence agent — webhook triggers, normalization/validation nodes, routing to internal services — over ERP-derived production data; contributed to a deployment the team reported cut recurring query turnaround from a working day to seconds.
- Implemented structured-output validation with retry and exception paths, API error handling and workflow test cases so malformed model or external-service responses were rejected rather than passed downstream.

Cardbyte

Automation Project Lead Intern

Jul 2025 – Aug 2025

Gurugram, India

- Worked within an existing Java/Selenium/TestNG/Maven framework, specifying test scenarios and then reviewing, debugging and modifying test scripts and Page Object classes to validate web and mobile UI.
- Built a UI element crawler (below) to automatically discover interactive elements across application pages in support of test-case authoring.

PROJECTS

Battery Characterization Suite — Python, SciPy, pandas, Streamlit, pytest

github.com/PaarthPatodia/battery-ecm-toolkit

- Built four interoperating tools extracting battery-model parameters from cyclor data: coulomb-counted SOC, OCV at relaxation, charge/discharge hysteresis, and a shared 1RC–5RC Thevenin ECM fitted across all pulse windows by bounded nonlinear least squares, with holdout scoring and per-parameter standard errors.
- Validated every analysis against a synthetic cell with known ground truth, recovering SOC to 0.02%, OCV to 0.9 mV and hysteresis to 0.53 mV against a 13–24 mV signal, and a known 2RC circuit to within 1.1% from data carrying 0.5 mV noise.
- Wrote a parser converting free-text DOE spreadsheets into executable current waveforms, and isolated colliding analysis backends into subprocesses behind a versioned CSV-plus-manifest contract, covered by integration tests running the real backends in CI.

UI Element Scraper — Java, Selenium, Maven

github.com/PaarthPatodia/ui-element-scraper

- Built a Java tool that signs into a web application and crawls it depth-first, discovering interactive controls — including ones reachable only inside modals, tabs and dropdowns — that a flat single-pass scan would miss.
- Designed a state-graph crawl strategy with independent termination bounds (depth cap, per-trigger visit set, element identity set, global budget) to guarantee termination on a cyclic, effectively unbounded UI graph, and emitted a structured element inventory with locator-quality scoring.